# Supplementary Materials for "A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology"

This supplementary materials includes proofs of technical results and additional numerical examples supporting conclusions from Section 4 the main manuscript.

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# 1 Proofs

## 1.1 Proof of Lemma 3.2

*Proof.* Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k}))}, f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. A similar technique can be applied to the second equation to obtain the result. □

## 1.2 Proof of Theorem 3.4

*Proof.* First, Lemma 3.2 allows us to write

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. We can rewrite $h_{j,k}(\cdot)$ into a form that will allow for specific computation involving Beta distributions. To see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F^{-1}_Y \circ U_{(j)}$ where $U_{(j)} \sim \mathrm{Beta}(j, n_1 - j + 1)$.

$$\begin{aligned}
&\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left\{ \int_x^{F^{-1}_X \circ I^{-1}_{r_{1,j}, N_1 - r_{1,j}+1} \circ I_{j, n_1 - j+1} \circ I^{-1}_{k, n_2 - k+1} \circ I_{r_{2,k}, N_2 - r_{2,k}+1} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ \int_x^{F^{-1}_X \circ I^{-1}_{r_{1,j}, N_1 - r_{1,j}+1} \circ I_{j, n_1 - j+1} \circ I^{-1}_{k, n_2 - k+1} \circ I_{r_{2,k}, N_2 - r_{2,k}+1} \circ F_X(x)} d\, I_{r_{1,j}, N_1 - r_{1,j}+1}(F_X(y)) \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j, n_1 - j+1} \circ I^{-1}_{k, n_2 - k+1} \circ I_{r_{2,k}, N_2 - r_{2,k}+1} \circ F_X(x) - I_{r_{1,j}, N_1 - r_{1,j}+1} \circ F_X(x) \right\}^+ f_{X_{2,(r_{2,k})}}(x)\, dx
\end{aligned}$$

$$= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ \; d\, I_{r_{2,k},N_2-r_{2,k}+1}(F_X(x))$$
$$= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt.$$

Condition (8) states that

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt$$
$$= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\} \; dt.$$

Thus,

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt$$
$$= \int_0^1 I_{j,n_1-j+1}(x) \; dI_{k,n_2-k+1}(x) - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x) \; dI_{r_{2,k},N_2-r_{2,k}+1}(x)$$
$$= \int_0^1 I_x(j, n_1-j+1) \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)} \; dx$$
$$- \int_0^1 I_x(r_{1,j}, N_1-r_{1,j}+1) \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(k, N_2-k+1)} \; dx$$
$$= \int_0^1 \sum_{p=j}^{n_1} \left( \begin{array}{c} n_1 \\ p \end{array} \right) x^p (1-x)^{n_1-p} \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)} \; dx$$
$$- \int_0^1 \sum_{p=r_{1,j}}^{N_1} \left( \begin{array}{c} N_1 \\ p \end{array} \right) x^p (1-x)^{N_1-p} \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2-r_{2,k}+1)} \; dx$$
$$= \sum_{p=j}^{n_1} \left( \begin{array}{c} n_1 \\ p \end{array} \right) \frac{B(k+p, n_2+n_1-p-k+1)}{B(j, n_2-k+1)}$$
$$- \sum_{p=r_{1,j}}^{N_1} \left( \begin{array}{c} N_1 \\ p \end{array} \right) \frac{B(r_{2,k}+p, N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.$$

□

## 1.3 Proof of Proposition 3.5

*Proof.* Specification of $1_i(X_{i,j} \geq X_{i,(N_i-n_i+1)})$ implies that $r_{i,j} = N_i + 1 - j$. When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $x \in (0,1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever $n_1$ or $N_1$ are increased [Askitis, 2021], or $n_2$ or $N_1$ are decreased. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned}
&I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\
&\implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\
&\implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t.
\end{aligned}$$

Technical condition (8) holds and the conclusion of Theorem 3.4 applies. □

## 1.4 Proof of Proposition 4.2

*Proof.* First, in combination with the Definition 4.1 and the assumption $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, we have

$$F_{Y_1}(x)^{\frac{n_1}{N_1}} \geq F_{Y_2}(x)^{\frac{n_2}{N_2}} \implies x \geq F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right).$$

By applying the Equation 7 from the Lemma 3.2 we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \mathrm{P}\left(F^{-1}_{Y_1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{-\infty}^{+\infty} \int_{F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x)
\end{aligned}$$

$$= \int_{-\infty}^{+\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_{2,(n_2)}}(x)$$
$$= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt.$$

Based on the Definition 4.1, we have

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\geq \int_0^1 F_{Y_2}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\leq \int_0^1 \left(t^{1/n_2} + \varepsilon\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}.
\end{aligned}$$

□

## 1.5 Proof of Theorem 4.3

*Proof.* Equation 7 from Lemma 3.2 yields

$$g_{n_1,n_2}(x) = F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right) = \Phi^{-1}\left(\Phi\left(\frac{x-\mu_2}{\sigma_2}\right)^{\frac{n_2 N_1}{N_2 n_1}}\right)\sigma_1 + \mu_1.$$

Let $A$ and $R$ be as defined, respectively, in (11) and (13). We have,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right)$$

$$
\begin{aligned}
&= \mathrm{P}\left(F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_A \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x) \\
&= \int_A F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) - \int_{A^c} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) + R.
\end{aligned}
$$

We continue with the left-hand side of the last equality,

$$
\begin{aligned}
&\int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) = \int_0^1 \left[F_{Y_1}\left(F_{Y_2}^{-1}(t^{1/n_2})\right)^{n_1} - t^{N_1/N_2}\right] dt \\
&\quad = \int_0^1 \left(\Phi\left(\frac{\Phi^{-1}(t^{1/n_2})\sigma_2 + \mu_2 - \mu_1}{\sigma_1}\right)\right)^{n_1} dt \; - \; \frac{N_2}{N_1 + N_2} \\
&\quad = \int_{-\infty}^{+\infty} \Phi\left(\frac{\sigma_2 x + \mu_2 - \mu_1}{\sigma_1}\right)^{n_1} \left(n_2 \Phi(x)^{n_2 - 1}\phi(x)\right) dx \; - \; \frac{N_2}{N_1 + N_2} \\
&\quad = n_2 \int_{-\infty}^{+\infty} \mathbb{P}\left( \begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2 - \mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2 - 1} - X \le 0 \end{array} \;\middle|\; X = x \right) \phi(x)\, dx \; - \; \frac{N_2}{N_1 + N_2} \\
&\quad = n_2\, \mathbb{P}\left( \begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2 - \mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2 - 1} - X \le 0 \end{array} \right) \; - \; \frac{N_2}{N_1 + N_2} \\
&\quad = n_2\, F_{\mathbf{W}}\left(\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, \frac{\mu_2 - \mu_1}{\sigma_1}, 0, \ldots, 0)\right) \; - \; \frac{N_2}{N_1 + N_2}.
\end{aligned}
$$

The distribution function $F_{\mathbf{W}}$ corresponds to random variable $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ where $\mathbf{\Sigma}$ is as defined in (14) and (15). The first $n_1$ elements of $\mathbf{w}$ is $\frac{\mu_2 - \mu_1}{\sigma_1}$ and the last $n_2 - 1$ elements are 0.

The bound on the remainder (17) follows from the argument in Section 4.2. □

## 1.6 Proof of Theorem 4.4

*Proof.*

$$\mathbf{F}_{\mathbf{W}}(\mathbf{w}) = \mathbb{P}\left( \begin{array}{l} Z_1 + \frac{\sigma_2}{\sigma_1} Z_0 \leq \frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, Z_{n_1} + \frac{\sigma_2}{\sigma_1} Z_0 \leq \frac{\mu_2 - \mu_1}{\sigma_1}, \\ Z_{n_1+1} + Z_0 \leq 0, \ldots, Z_{n_1+n_2-1} + Z_0 \leq 0 \end{array} \right)$$
$$= \mathbb{P}\left( \begin{array}{l} Z_1 \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \ldots, Z_{n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \\ Z_{n_1+1} \leq -Z_0, \ldots, Z_{n_1+n_2-1} \leq -Z_0 \end{array} \right)$$

Define $M_{n_1} = \max\{Z_1, Z_2, \ldots, Z_{n_1}\}$, $M_{n_2-1} = \max\{Z_{n_1+1}, Z_{n_1+2}, \ldots, Z_{n_1+n_2-1}\}$, then the CDF of $\mathbf{W}$ essentially computes the joint probability:

$$P(M_{n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0 \text{ and } M_{n_2-1} \leq -Z_0)$$

From extreme value theory, as $n_1 \to \infty$, $M_{n_1}$ sharply concentrates around $\sqrt{2 \ln n_1}$ [Embrechts et al., 2013].

Substitute this into the first inequality:

$$\sqrt{2 \ln n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0 \implies Z_0 \leq \frac{\frac{\mu_2 - \mu_1}{\sigma_1} - \sqrt{2 \ln n_1}}{\frac{\sigma_2}{\sigma_1}}$$

As $n_1$ grows, $\frac{\frac{\mu_2 - \mu_1}{\sigma_1} - \sqrt{2 \ln n_1}}{\frac{\sigma_2}{\sigma_1}}$ becomes a very large negative number. This forces the random variable $Z_0$ deep into its left tail.

Similarly, as $n_2 \to \infty$, $M_{n_2-1}$ sharply concentrates around $\sqrt{2 \ln(n_2 - 1)}$ [Embrechts et al., 2013].

Substitute this into the second inequality:

$$\sqrt{2 \ln(n_2 - 1)} \leq -Z_0 \implies Z_0 \leq -\sqrt{2 \ln(n_2 - 1)}$$

As $n_2$ grows, $-\sqrt{2\ln(n_2-1)}$ becomes a very large negative number. This forces the random variable $Z_0$ deep into its left tail.

Then the CDF of $\mathbf{W}$ essentially computes the probability:

$$\mathbf{P}(Z_0 \leq \frac{\frac{\mu_2-\mu_1}{\sigma_1} - \sqrt{2\ln n_1}}{\frac{\sigma_2}{\sigma_1}}, Z_0 \leq -\sqrt{2\ln(n_2-1)}),$$

as $n_1, n_2 \to \infty$.

This probability is dominated entirely by the tail of $Z_0$. We can use the Mill's ratio [Mills, 1926] and apply it to standard Gaussian tail approximation, $\Phi(-x) \sim \frac{\phi(x)}{x}$ as $x \to \infty$.

Since

$$\lim_{n_1,n_2\to\infty} \frac{\ln n_2}{\ln n_1} > \left(\frac{\sigma_1}{\sigma_2}\right)^2,$$

the tail behavior of $Z_0$ is dominated by $n_2$, then

$$F_{\mathbf{W}}(\mathbf{w}) \approx \Phi(-\sqrt{2\ln(n_2-1)}) \sim \frac{\phi(\sqrt{2\ln(n_2-1)})}{\sqrt{2\ln(n_2-1)}} = \frac{1}{\sqrt{4\pi\ln(n_2-1)}}\left(\frac{1}{n_2-1} + o(\frac{1}{n_2})\right)$$

Therefore, the asymptotic paradox probability in equation (16):

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \mid \hat{\mu_1}, \hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right) \\
&\quad - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \mid \mu_1, \mu_2, \sigma_1, \sigma_2\right) \\
&= n_2(F_{\mathbf{W}}(\mathbf{w}) - F_{\hat{\mathbf{W}}}(\hat{\mathbf{w}})) \to 0
\end{aligned}$$

as $n_1, n_2 \to \infty$.

□

# 2 Simulations

## 2.1 Simulation Results for Theorem 4.3

We now show the simulation results of our Normal Distribution example under different configurations of $(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$. We verify Theorem 4.3 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for $Y_1$ and $Y_2$ are specified along with sample sizes $n_i$, $N_i$, $i = 1, 2$. We will specify a Monte Carlo sample size of 2000. For iteration $m$ of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (2) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The theoretical results are computed from the Theorem 4.3 using mvtnorm package [Genz and Bretz, 2009] in R.

The results are tabulated in Table 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

| $n_1$ | $n_2$ | $N_1$ | $N_2$ | $\mu_1$ | $\mu_2$ | $\sigma_1$ | $\sigma_2$ | Simulation Prob | Theoretical Prob |
|---|---|---|---|---|---|---|---|---|---|
| 10 | 40 | 400 | 600 | -1 | 1 | 1.0 | 3 | 0.4000 | 0.4000099 |
| 10 | 20 | 400 | 600 | 0 | 0 | 1.0 | 3 | 0.3970 | 0.3970305 |
| 10 | 40 | 400 | 300 | -1 | 1 | 1.0 | 3 | 0.5715 | 0.5714174 |
| 10 | 20 | 400 | 600 | 0 | 1 | 1.5 | 2 | 0.3645 | 0.3647783 |
| 10 | 40 | 400 | 600 | 0 | 1 | 1.0 | 3 | 0.3995 | 0.3999245 |
| 20 | 40 | 400 | 600 | 0 | 1 | 1.0 | 3 | 0.3995 | 0.3999803 |
| 10 | 40 | 400 | 600 | -1 | 0 | 1.5 | 3 | 0.4000 | 0.3995149 |
| 20 | 40 | 400 | 600 | 0 | 0 | 1.5 | 2 | 0.2970 | 0.2962431 |
| 20 | 40 | 400 | 600 | 0 | 0 | 1.0 | 3 | 0.3750 | 0.3740938 |
| 10 | 20 | 400 | 600 | -1 | 1 | 1.0 | 3 | 0.3990 | 0.4000059 |

Table 1: Simulation Results for Normal Example.

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